



COURSE SYLLABUS						
Cours e Code	COMP 027	Course Title	Mobile App Development		Course Credit	3
Course Description		This course introduces students to the fundamentals and advanced concepts of mobile app development for Android and cross-platform systems. It emphasizes practical application through hands-on projects, covering UI/UX, data handling, networking, APIs, security, and deployment. Students will design, build, and deploy functional mobile apps by the end of the course.				
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				

WORLD UNIVERSITY RANKINGS

2025

7th in the Philippines
92nd in Southeast Asia
541-560 In Asia

WURI

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL PROMOTION

QS STARS

5 stars

Employability
Online Learning
Internationalization

4 stars

Teaching
Academic Development
Inclusiveness

3 stars

Social Responsibility
Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education

Impact Rankings 2024

DEBUTS AT 1501+



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





6. **Leadership and Organizational Skills.** Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills.
7. **Personal and Professional Ethics.** Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances.
8. **Resilience and Agility.** Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment.
9. **National and Global Responsiveness.** Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.

COLLEGE GOALS AND OBJECTIVES

- The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals:
1. Offer curricula which are relevant and responsive to the changing needs of the industry and society;
 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development.
 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization.
 4. Provide state-of-the-arts facilities of learning to optimize student development;
 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society;
 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development;
 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development;
 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students;
 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals;
 - 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and,
 - 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)

Based on CMO

Apply knowledge of computing, science, and mathematics appropriate to the disciplin





	Analyze complex problems, and identify and define the computing requirements appropriate to its solution												
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.												
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.												

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DEBUTS AT 1501+



	Integrate IT-based solutions into the user environment effectively.													
	Assist in the creation of an effective IT project plan.													
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession													
	Understand best practices and standards and their applications.													
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.													
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.													
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.													
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.													
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development													

COURSE LEARNING OUTCOMES (CLO)	Alignment to PLOs							
	1	2	3	4	5	6	7	8

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- Inclusiveness

3 STARS
- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

DEBUTS AT 1501+



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OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/ Topics	Instructional Delivery Design		Assessment		Alignmen t to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)			1	2	3	4
					Face-to-Face	FLTAs				
1	ORIENTATION & GITHUB SETUP	Course overview, objectives, grading policy, and project guidelines Git & GitHub setup: accounts, repositories, commits, branches, pull requests Install Android Studio / Flutter SDK / VS Code			Create a repo named MobileDev-StudentName and upload a simple README.md with self-introduction and environment setup screenshot.		X			
2	INTRODUCTION TO MOBILE DEVELOPMENT	Mobile OS architectures (Android vs iOS) Native vs Hybrid vs Cross-platform development Setting up a sample project in Android Studio or Flutter			Push a simple “Hello Mobile Dev” project (splash screen with text).					

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3	PROGRAMMING FOUNDATIONS FOR MOBILE APPS	Refresher: Java/Kotlin/Dart basics App lifecycle and activity management			Create a multi-activity Hello World app with at least 2 screens and navigation. Push source code.					
4	USER INTERFACE DESIGN	UI/UX principles in mobile apps Layouts, views, widgets, themes, material design			Build a multi-screen UI (Login, Home, Profile). Commit design screenshots and project files.					
5	USER INPUT & EVENT HANDLING	Forms, gestures, sensors Input validation and error handling			Create a registration form app with field validation. Push code and test screenshots.					
6	DATA STORAGE	Shared Preferences, local storage, SQLite			Develop a note-taking app (add, edit, delete). Upload project repo.					
7	NETWORKING & APIs	REST APIs, JSON parsing, Retrofit/HTTP			Create a weather/news app using a free public API. Upload code and sample run.					
8	ADVANCED UI AND NAVIGATION	Advanced UI and Navigation			Build an app with bottom navigation bar & tabbed layout . Push to repo.					



9	MIDTERM WRITTEN EXAMINATION & PROJECT PITCH									
10	CLOUD & FIREBASE INTEGRATION	Firestore, Firebase Auth, notifications			Build a simple chat app using Firebase backend. Upload repo and screenshots.					
11	DEVICE FEATURES & SENSORS	GPS, camera, microphone, accelerometer			Create a location tracker app showing user's live location on map. Push to repo.					
12	SECURITY IN MOBILE APPS	Authentication, encryption, secure storage OWASP Mobile Top 10 basics			Build a secure login system (Firestore/Auth). Upload repo with security notes.					
13	PERFORMANCE & OPTIMIZATION	Memory and battery optimization Debugging & profiling tools			Take an earlier app, profile & optimize it (reduce lag, improve UI). Commit before/after changes.					
14	CROSS-PLATFORM DEVELOPMENT	Intro to Flutter/React Native Compare native vs cross-platform			Build a cross-platform to-do list app in Flutter/React Native. Upload to repo.					
15	PUBLISHING & DEPLOYMENT	Preparing apps for release			Generate a signed APK or release build and upload final release file + instructions in README.					
16	CAPSTONE PROJECT DEVELOPMENT	Full implementation and debugging of final project			Push beta version of final project to repo.					
17	FINAL PROJECT DEFENCE / DEMO DAY									

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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies ● Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		● Quizzes						
18	FINAL PROJECT DEFENCE/DEMO										
LIST OF REFERENCES BOOKS: <											





CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic

dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must

Intellectual Property and Copyright.

Original materials created for this online class is covered by university policy. Third-party (third-party) instructional materials (e.g., YouTube videos) might be under copyright while others (e.g., some assigned readings) might be subject to fair use. All





	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Quizzes		Quizzes	
Long Test		Long Test	
Recitation and Attendance		Recitation and Attendance Research	
Research Work		Work	
Assignment	30%	Assignment	30%
Projects		Projects	
Exercises		Exercises	
Midterm Examinations		Final Examinations	100%
	100%		

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.
COURSE REQUIREMENT/S	GRADING SYSTEM



- Making a Learning Plan in Teaching Prose (focusing on Completeness, Appropriate Approach Used, and Facilitative Process) - Teaching Demonstration in Prose or poetry (focusing on Teacher- Like Simulation, Preparation, Classroom Management, Learning Environment, and Interest and Diligence in Teaching Poetry)	FINAL GRADE = (FIRST GRADING + SECOND GRADING) / 2
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CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section:		Name of Faculty:	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester:	E-Mail Address:	



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Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
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